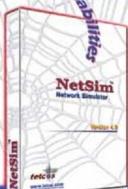


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Every other country in the world pays a high rate for such solar power fed into the grid and this makes such investment very well rewarding.

have proliferated. There are a variety of means to tilt the panels toward the sun to generate more electricity.

Currently, commercially available silicon-based solar panels (called PV solar panels) are made from solar cells encased in a special type of toughened glass. Silicon solar modules have been in the field for more than 50 years and perform quite predictably. These are guaranteed for 25 years of field life but the power yield drops about 0.6 per cent a year. One can use monocrystalline (made from a single crystal) or polycrystalline (made from multiple crystals) panels. Mono-crystalline panels are a bit more efficient but the cost per watt is almost the same.

Copper-indium gallium-diselenide (CIGS) panels may become a preferred type for commercial rooftop projects in another five years. These have the potential to deliver reasonable efficiencies at a lower cost than traditional crystalline panels. However, the cost per watt may not necessarily go down, only the panel size per watt will drop. Note that today solar panels (depending on the brand) are bankable. That is, banks will loan capital for their purchase.

How are solar panels installed on rooftop?

Many solar system installers and owners have had good experiences with penetrating anchoring of panel structures if done scientifically and with care. Non-anchored installation systems are also possible but need careful design to withstand heavy winds. Such systems are designed to connect the solar power system to a roof using weights, rather than fasteners that must be anchored in the roof. Solar installation companies, often called integrators, can complete a small rooftop project in several weeks.

Before signing a contract with an integrator to install a solar system, consider the following issues:

1. Roof condition—if you have an old roof that needs significant maintenance or replacement, you may want to do that before installing the solar array.
2. Space availability—solar power projects work best on flat roofs without obstructions.
3. Weight load—some roofs are not designed to hold much additional weight. Ascertain the acceptable weight you can add to your roof before signing a contract.

Go solar!

When you have a rooftop PV solar plant, you will become a power manufacturer. In the long run, the cost of traditional fossil fuel electricity will increase, while the cost of solar electricity will decrease. It is still murky to determine as to when solar power will cost less than fossil fuel electricity but it may happen soon since fossil fuels are becoming more and more expensive and depleting at a fast pace. Based on the current prices and assuming that one takes advantage of the 80 per cent depreciation permitted on such investment in the first year, the cost of power per unit, kWh, from a well maintained solar plant will be less than Rs 8 per watt for plant of any capacity between 5 kW and 1 MW. (I will be able to further elaborate this for those who want to know.)

As the price of coal increases and the cost of building new coal and oil and nuclear power plants continues to climb, renewable-energy alternatives like solar will become much more attractive to everyone. In any case, if the government and successive governments honour their commitment to pay for your solar energy fed into the grid at Rs 17, you have a profitable deal! ●

The author is ex-chairman, Electronics Commission, government of India, and a former advisor to late Rajiv Gandhi